

Suppose both X and Y are discrete, $(x, y) \in \mathcal{X} \times \mathcal{Y}$
 (x, y)

The probability mass function of (X, Y) satisfies

$$p_{X,Y}(x_i, y_j) = \mathbb{P}(X=x_i, Y=y_j), \quad (x_i, y_j) \in \mathcal{X} \times \mathcal{Y}$$

$$(i) \quad p_{X,Y}(x_i, y_j) \geq 0 \text{ for all } (x_i, y_j)$$

$$(ii) \quad \sum_{x_i} \sum_{y_j} p_{X,Y}(x_i, y_j) = 1$$

Ex:- Suppose a car showroom has 10 cars of a brand out of which 5 are good (G), 2 have defective transmissions (DT), 3 have defective steering (DS). If 2 cars are selected at random and X denotes the no of cars with DT and Y denote the no of cars with DS, find the joint PMF of (X, Y) .

Soln: X : no of cars with DT $0, 1, 2$

Y : no of cars with DS. $\rightarrow 0, 1, 2$

$$p_{X,Y}(0,0) = \frac{\binom{5}{2}}{\binom{10}{2}} = \frac{10}{45}$$

$$p_{X,Y}(0,1) = \frac{\binom{5}{1} \binom{3}{1}}{\binom{10}{2}} = \frac{15}{45}$$

$\vdots$

PMF of the RV X

		0	1	2	
	X \ Y	0	1	2	$P_X(x)$
→	0	$\frac{10}{45}$	$\frac{15}{45}$	$\frac{3}{45}$	$\frac{28}{45}$
→	1	$\frac{10}{45}$	$\frac{6}{45}$	0	$\frac{16}{45}$
→	2	$\frac{1}{45}$	0	0	$\frac{1}{45}$
→	$P_Y(y)$	$\frac{21}{45}$	$\frac{21}{45}$	$\frac{3}{45}$	1

PMF of the RV Y.

$$P(X \leq 1, Y \leq 1) = P(X=0, Y=0) + P(X=0, Y=1) + P(X=1, Y=0) + P(X=1, Y=1)$$

$$= \frac{10}{45} + \frac{15}{45} + \frac{10}{45} + \frac{6}{45}$$

$$= \frac{41}{45}$$

$$P(X < 2) = P(X=0) + P(X=1)$$

$$= \frac{28}{45} + \frac{16}{45} = \frac{44}{45}$$

- The marginal pmf of the RV X is

$$P_X(x_i) = \sum_{y_j} P_{X,Y}(x_i, y_j), \quad x_i \in \mathcal{X}$$

The marginal pmf of the RV Y is

$$P_Y(y_j) = \sum_{x_i} P_{X,Y}(x_i, y_j), \quad y_j \in \mathcal{Y}$$

In the above example, the marginal pmf of X is given by

$$P(X=0) = \frac{28}{45}, \quad P(X=1) = \frac{16}{45}, \quad P(X=2) = \frac{1}{45}$$

and the marginal pmf of Y is given by

$$P(Y=0) = \frac{21}{45}, \quad P(Y=1) = \frac{21}{45}, \quad P(Y=2) = \frac{3}{45}$$

- The conditional pmf of X given $Y=y_j$ is

$$p_{X|Y=y_j}(x_i) = \frac{p_{XY}(x_i, y_j)}{p_Y(y_j)}, \quad x_i \in \mathcal{X}$$

- The CDF of the RV (X, Y) is

$$\underline{F_{X,Y}(x,y) = P(X \leq x, Y \leq y), \quad (x,y) \in \mathbb{R}^2}$$

$$(X, Y) \quad f_{X,Y}(x,y)$$

$$(i) \quad f_{X,Y}(x,y) \geq 0 \quad \forall (x,y) \in \mathbb{R}^2$$

$$(ii) \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx dy = 1$$

$$f(x,y) = e^{-(x+y)}, \quad 0 < x < \infty, \quad 0 < y < \infty$$
$$= 0, \quad \text{otherwise}$$

$$\begin{aligned}
\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy &= \int_0^{\infty} \int_0^{\infty} e^{-(x+y)} dx dy \\
&= \int_0^{\infty} e^{-x} dx \times \int_0^{\infty} e^{-y} dy \\
&= [-e^{-x}]_0^{\infty} \times [-e^{-y}]_0^{\infty} = 1
\end{aligned}$$